

Exercise 05

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Written Part

1. What are the differences between the data structures Tree and Graph? Explain using the mathematical definitions of tree and graph.

Mathematical definitions:

- **Graph:** A graph $G = (V, E)$ consists of a set of vertices V and a set of edges $E \subseteq V \times V$.
 - Edges can be directed or undirected. (so may or may not be connected)
 - Graphs can have cycles, multiple components, and any structure.
- **Tree:** A tree is a special type of graph:
 - A **connected, acyclic** undirected graph.
 - For a tree with n vertices, it has exactly $n - 1$ edges.

2. What is the maximum number of edges that a simple disconnected undirected graph with 8 vertices can have? Provide a proof to your answer.

A disconnected graph has at least two components.

Thus, to maximize edges while keeping the graph disconnected, we split vertices into two components.

Let the components have k and $8-k$ vertices. Each component can be a complete graph, so the number of edges:

$$E_{\max} = \binom{k}{2} + \binom{8-k}{2} = \frac{k(k-1)}{2} + \frac{(8-k)(7-k)}{2}.$$

- To maximize $E_{\max}$ we take $k=1$ or $k=7$ (most unbalanced split):

$$E_{\max} = \binom{1}{2} + \binom{7}{2} = 0 + 21 = 21.$$

Answer: Maximum edges = 21

3. There are 23 routers in university building. Is it possible to connect them with cables such that each router is connected to exactly 5 others? Explain the reason.

- **Condition:** Each router has degree 5 $\rightarrow$ the sum of all degrees = $23 \times 5 = 115$
- **Handshaking lemma:** In any graph, the sum of degrees must be **even**, because each edge contributes 2 to the sum.

$$\text{Sum of degrees} = 2 \cdot |E| \quad \Rightarrow \quad 115 \text{ is odd, impossible.}$$

Answer: Impossible, because 115 is odd, violating the handshaking lemma.

4. In graph theory, a bipartite graph is a graph whose vertices can be divided into two disjoint and independent sets U and V , where every edge connects a vertex in U to one in V .

Prove that the size of a bipartite graph of order (number of vertices) n is at most at $\frac{n^2}{4}$

Problem: Show that a bipartite graph with (n) vertices has at most $(\frac{n^2}{4})$ edges.

Step 1: Define a bipartite graph

- Let $G = (V, E)$ be a bipartite graph.
 - Its vertex set V can be divided into **two disjoint sets** U and V , i.e., $V = U \cup V$ and $U \cap V = \emptyset$.
 - By definition, all edges connect a vertex in U to a vertex in V
 - Let $|U| = u$ and $|V| = v$. Then $u + v = n$, the total number of vertices.
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Step 2: Number of edges

- Maximum edges occur when **every vertex in U is connected to every vertex in V**.
- In that case, the total number of edges is: $|E|_{\max} = u \cdot v$, because each of the u vertices in U connects to all v vertices in V .

Step 3: Maximize $u \cdot v$ given $u+v = n$

- Substitute $v = n - u$ into $u \cdot v$:

$$|E|_{\max} = u \cdot (n - u) = un - u^2$$

- Now we have a quadratic function $f(u) = -u^2 + nu$
- A quadratic function $f(u) = -u^2 + nu$ reaches its maximum at $u = -\frac{b}{2a}$ where $(a = -1, b = n)$:

$$u_{\max} = -\frac{n}{2 \cdot (-1)} = \frac{n}{2}.$$

Step 4: Compute

- If $u = v = \frac{n}{2}$ (assuming n is even), then:

$$|E|_{\max} = u \cdot v = \frac{n}{2} \cdot \frac{n}{2} = \frac{n^2}{4}.$$

Therefore, a bipartite graph with n vertices has at most $\frac{n^2}{4}$ edges.

Step 5: Remarks

- If n is odd, then one part has $\frac{n-1}{2}$ vertices and the other has $\frac{n+1}{2}$ vertices. Then the product is:

$$\frac{n-1}{2} \cdot \frac{n+1}{2} = \frac{n^2-1}{4} < \frac{n^2}{4}.$$

So the upper bound $\frac{n^2}{4}$ always holds.

5. How does the Kronecker product of a complete graph with itself look like?

The Kronecker product of a complete graph K_n with itself, results in a graph with n^2 vertices, each labeled as an ordered pair (i,j) where $i,j \in \{1,\dots,n\}$.

In this product graph, **two vertices (i,j) and (k,l) are connected if and only if $i \neq k$ and $j \neq l$** . This means adjacency occurs only when both coordinates differ — a stricter condition than in the original complete graph.

Visually,

- $n \times n$ grid
- each vertex connects to all others outside its row and column.
- Example: In $K_3 \otimes K_3$, each vertex connects to $(3 - 1)^2 = 4$ others.

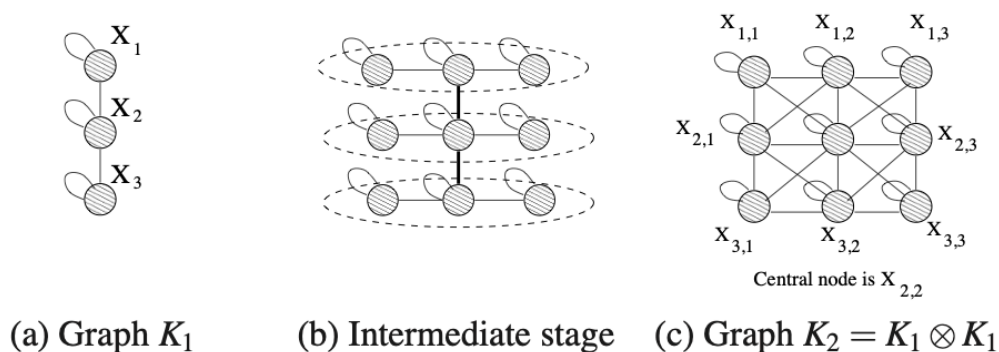


Figure 1 (a,b, c) from <https://cs.stanford.edu/~jure/pubs/kronecker-jmlr10.pdf>

The figure below illustrates the **Kronecker product of a graph with itself**, though the seed graph K_1 (left) is not a complete graph, but a 3-node path-like graph.

Nevertheless, the process is the same: the Kronecker product expands the graph into a larger structure where connections follow the Kronecker rule of adjacency.

In the case of a complete graph K_n , the result would be a grid-like graph where each vertex connects to all vertices not sharing its row or column, producing a pattern more densely connected than shown here, but following the same underlying logic.